Middlesex University Dubai

CARA

Companion Assistive Robot for the Aged

A Low-Cost GSM-Based Companion Assistive Robot for
Elderly Care and Human–Robot Interaction Research in Developing Countries

Master of Science in Robotics

Jayashanka Anushan
Student ID: M01037028
Supervisor: Prof. Judhi
Middlesex University Dubai | July 2026

Declaration of Originality

I hereby declare that the dissertation entitled " CARA — A Low-Cost GSM-Based Companion Assistive Robot for Elderly Care and Human–Robot Interaction Research in Developing Countries " submitted in partial fulfilment of the requirements for the degree of Master of Science in Robotics at Middlesex University Dubai is entirely my own work.

The work contained in this dissertation has not been submitted for any other degree or professional qualification. All sources cited in this dissertation have been properly acknowledged in accordance with the IEEE referencing system.

I confirm that the use of artificial intelligence tools in this work is fully disclosed in the AI Usage Statement provided in the appendices. All sources cited in this dissertation have been properly acknowledged, and the research contributions including design framework, software, and testing methodology are the author's own work.

Signature	:	______________________________
Name	:	Jayashanka Anushan (M01037028)
Date	:	

Abstract

The global population is ageing rapidly, with developing countries experiencing particularly acute challenges due to economic migration resulting in millions of elderly people living alone without consistent daily care. Existing assistive robotic solutions are designed for high-income, infrastructure-rich environments and are financially and technologically inaccessible to most families in developing regions. This dissertation presents CARA (Companion Assistive Robot for the Aged), a low-cost GSM-based companion assistive robot designed specifically for elderly care in developing countries.

The primary aim of this research was to design, build, and evaluate a mobile assistive robot capable of providing safe monitoring, basic health support, and family connectivity without relying on continuous internet access or active elderly user interaction. The system integrates an NVIDIA Jetson Nano for AI-based fall detection using MediaPipe pose estimation, an Arduino Mega for real-time control and sensor management, and an A7670C GSM module for Wi-Fi-independent emergency alerting.

Fifteen functional features were implemented and evaluated, including autonomous indoor navigation with obstacle avoidance, AI-powered fall detection, GSM-based SMS emergency alerts, medication reminders with RFID confirmation, live camera streaming for family remote monitoring, and companion interaction features including greetings and status display.

Evaluation was conducted through standardized trial protocols in a controlled laboratory environment. Fall detection achieved 80% accuracy across 20 standardized trials with a false positive rate below 2 per hour. GSM emergency alerts were delivered within 30 seconds of fall confirmation across 15 test trials. Navigation achieved collision-free operation in over 80% of trials. The complete 15-feature system was evaluated against defined user requirements.

The research contributes a systematic integration framework combining AI computer vision, microcontroller control, and cellular communication for low-resource elderly care environments. Empirical guidelines for optimizing MediaPipe fall detection on low-cost embedded hardware are provided. A zero-interaction passive monitoring architecture suitable for elderly populations with low digital literacy is demonstrated. A template for adapting high-income assistive technologies to developing world contexts is contributed.

CARA Version 1.0 is presented as a research platform and proof-of-concept prototype. While technical feasibility is demonstrated, limitations including single-subject testing, laboratory-only evaluation, and 80% fall detection accuracy (below commercial standards) are acknowledged. These limitations provide clear directions for future development including elderly user testing, extended environmental trials, and accuracy optimization toward commercial deployment.

Keywords: Assistive Robotics, Elderly Care, Fall Detection, GSM Communication, Human-Robot Interaction, Developing Countries, MediaPipe, Ambient Assisted Living

Acknowledgements

I express my sincere gratitude to my supervisor, Prof. Judhi, for guidance throughout this research. Your expertise in robotics and systems engineering was invaluable in shaping this dissertation from an initial concept to a complete prototype.

I thank Middlesex University Dubai for providing the research facilities, laboratory space, and academic framework that enabled this work. The robotics laboratory resources were essential for implementation and testing.

I acknowledge the support of colleagues and fellow researchers who provided feedback during weekly progress reviews. Your insights helped identify technical challenges and refine the system architecture.

I thank the online open-source communities whose documentation and shared code accelerated the implementation phase. In particular, the MediaPipe, ROS, and Arduino developer communities provided foundational resources that made this project achievable within the constrained timeframe.

I express appreciation to my family for their patience and encouragement throughout this demanding process. Your support sustained me through the inevitable challenges of hardware debugging and system integration.

Finally, I dedicate this work to the elderly population of Sri Lanka and developing countries worldwide who face daily challenges with limited technological support. It is my hope that this research contributes, however modestly, to improving their quality of life and safety.

Middlesex University Dubai

Table of Contents

List of figures

List of Tables

1). Chapter: Introduction

1.1). Background and Motivation

Many developing countries are experiencing rapid demographic aging, accompanied by large-scale economic migration of working-age adults, particularly in Asia [1]. In Sri Lanka, the proportion of the population aged 60 and above is increasing sharply [2, 3] , and a significant number of young adults are migrating abroad for employment opportunities. As a result, older people are increasingly living alone in their family homes without consistent daily supervision or immediate care [4].

Older adults living alone face higher risks of falls, medication non-adherence, delayed medical response in emergencies, and social isolation [5]. These risks are further exacerbated by infrastructure constraints common in rural and semi-urban settings, including unreliable internet connectivity, limited access to formal care services, and low digital literacy among older populations [6]. At the same time, family members living abroad may experience ongoing anxiety due to the inability to regularly monitor their parents' well-being.

Assistive robotics and smart home technologies hold promise in addressing the challenges of elder care [7]. However, existing solutions are mainly designed for high-income, infrastructure-rich environments and are financially inaccessible to many families in developing regions [8, 9]. This creates a critical mismatch between where technological innovation is occurring and where the need is most urgent [10]. The motivation for this research is to bridge this gap by exploring how low-cost, infrastructure-independent assistive robotics can support older people living alone in developing countries [11].

1.2). Research Problem

Despite significant advances in assistive robotics and environmental assisted living technologies, current solutions for elder care in developing countries are largely inadequate. Commercially available assistive robots are expensive, require a stable internet connection, and assume a level of technological familiarity that most elderly users do not possess. Wearable safety devices and smart home systems similarly rely on user compliance, continuous charging, and reliable network infrastructure, which are often unrealistic in rural or resource-constrained environments.

As a result, older people living alone in such contexts are underserved by existing technological interventions. There is currently a shortage of affordable, autonomous, and context-aware robotic systems specifically designed for homes with limited infrastructure, low digital literacy, and limited financial resources. This research addresses the problem of how to design and evaluate a low-cost assistive robot that can provide safe monitoring, basic health support, and family connection without relying on continuous internet access or active user interaction.

1.3). Aim and Objectives

The aim of this dissertation is to design, build, and evaluate CARA as a research platform for studying humanoid social interaction, assistive behavior, and autonomous perception for elderly people living alone in developing countries, with specific application to the Sri Lanka context.

<u>This aim is pursued through the following five objectives:</u>

1. To design and implement an autonomous indoor navigation system using onboard camera-based, and ultrasonic sensors for obstacle avoidance, achieving collision-free navigation in at least 80% of trials in a single-floor indoor environment.
2. To develop and validate an AI fall detection system using MediaPipe pose estimation on NVIDIA Jetson Nano, achieving minimum 80% detection accuracy with a false positive rate below 2 per hour across 20 standardized trials.
3. To implement a GSM-based emergency alerting system delivering SMS notifications within 30 seconds of fall detection, operating without Wi-Fi dependency, validated across 15 test trials.
4. To design and build a health management subsystem including scheduled medication reminders, RFID-based confirmation, and automated family SMS alerts for missed doses.
5. To evaluate the complete 15-feature CARA system against defined user requirements using structured testing protocols and critically analyze performance against each stated objective.

1.4). Research Questions

This dissertation is guided by the following research questions:

- How can a low-cost mobile robot be designed to provide meaningful assistive support for elderly individuals living alone in developing-world home environments?
- To what extent can onboard computer vision deployed on low-cost hardware achieve reliable fall detection in real-world domestic settings?
- Can GSM-based communication provide a practical alternative to internet-dependent smart care systems for emergency alerts and family reassurance?
- What are the practical limitations and performance trade-offs of implementing multiple assistive features within a strict component cost budget?

– How can multimodal HRI be effectively studied in low-cost, GSM-based robotic platforms in developing world contexts?

1.5). Scope and Limitations

This research focuses on the design and prototyping of a single mobile assistive robot platform for home use. This scope includes safety monitoring (fall detection and emergency alerts), basic health management support (medication reminders and compliance verification), family connectivity features, and simple assistant interaction mechanisms.

The evaluation is limited to technical feasibility, functional performance, and initial usability testing in a controlled home-like environment. Long-term clinical validation, large-scale user studies with elderly participants, and formal medical outcome assessments are beyond the scope of this thesis. In addition, the system is not intended to replace human care or medical professionals but is intended to provide additional support and passive monitoring.

Hardware and algorithm limitations inherent to low-cost embedded platforms may limit accuracy, processing speed, and robustness. The study also does not address regulatory approval, commercial deployment, or large-scale manufacturing challenges that remain for future work.

Navigation is currently limited to a single-floor, indoor environment. PIR sensors are retained for presence detection, but do not provide position information. Multi-room mapping relies on low-cost visual SLAM using the camera and wheel odometry. PIR sensors are retained for presence detection only.

This dissertation is concerned with the technical design, implementation, and laboratory evaluation of CARA Version 1.0. The following scope boundaries apply:

Within scope:

- All 15 features as specified in the system design chapter
- Laboratory-based evaluation using standardized trial protocols
- Single-floor, single-room-per-test indoor environment
- Single user (researcher) as test subject for all trials
- GSM-based communication within UAE network coverage

Outside scope:

- Deployment in actual Sri Lanka home environments
- Testing with real elderly participants (ethical clearance not obtained for human subjects)
- Multi-floor navigation or outdoor operation

– Sinhala or Tamil language interaction
– Full clinical validation or medical device certification
– Mass production or commercial viability analysis

1.6). Dissertation Structure

This dissertation is organized as follows.

- Chapter 2 reviews the existing literature across eight areas relevant to CARA, culminating in a clear identification of the research gap.
- Chapter 3 presents the complete system design, including all hardware and software architecture decisions.
- Chapter 4 describes the implementation process for all 15 features, including challenges encountered and solutions applied.
- Chapter 5 presents the formal evaluation framework and results across all 15 features, with statistical analysis and comparison to the stated objectives.
- Chapter 6 concludes the dissertation with a summary of contributions, honest discussion of limitations, and directions for future work.

2). Chapter: Literature Review

2.1). Ageing and the Elderly Care Crisis

Sri Lanka is currently experiencing one of the most rapid demographic transitions in South Asia. The proportion of people aged 60 years and above increased from 12.4% of the population in 2012 and is projected to reach nearly one quarter of the population by 2041 [12]. Ageing is associated with increased limitations in activities of daily living (ADL) and instrumental activities of daily living (IADL), reducing independence and increasing the demand for long-term care services [13]. The share of the population aged 65 years and above reached approximately 17% in 2022, indicating a rapid demographic shift that places significant pressure on healthcare and social support systems [14].

According to recent census estimates, the elderly population in Sri Lanka increased from 12% in 2012 to approximately 18% in 2024 and is expected to reach 25% by 2040 [15]. The rising old-age dependency ratio indicates that a smaller working population must support a growing elderly population, posing economic and social challenges [16].

2.2). Assistive Robot Landscape

PARO, a robotic therapeutic seal, has been shown to improve emotional well-being and reduce agitation among elderly patients with dementia, although its high cost and supervised operation limit widespread deployment [17]. Pepper, a humanoid social robot, supports elderly users through greetings, conversations, and reminders, fostering social engagement and reducing feelings of isolation, though its reliance on WiFi limits use in connectivity-constrained settings [18]. Care-O-Bot 4 exemplifies a multi-functional home assistant capable of autonomous navigation and task assistance, though deployment is limited by cost and infrastructure requirements [19].

ElliQ, a socially assistive robot, engages elderly users with health reminders, conversation, and social prompts, emphasizing intuitive human–robot interaction, though its stationary design limits physical assistance [20]. Systematic reviews indicate that socially assistive robots can effectively reduce loneliness and stimulate cognitive activity among elderly users, although barriers such as cost, cultural acceptance, and the need for supervision remain significant [21]. Despite demonstrated benefits, most assistive robots remain inaccessible for low-resource settings, highlighting a critical gap in affordable, culturally adaptable robotic solutions [22].

Table 1 Comparison of Existing Assistive Robots (Author developed)

Robot	Developer	Cost (USD)	Key Features	Key Limitation for CARA Context
PARO	AIST, Japan	$5,000+	Companion seal, emotion response	No fall detection, no navigation, no alerting
PEPPER	SoftBank	$15,000+	Social interaction, voice	WiFi required, designed for public spaces
Care-O-Bot	Fraunhofer	$100,000+	Manipulation, navigation	Research prototype, not commercially available
ElliQ	Intuition Robotics	$250/month	AI companion, voice, screen	WiFi required, English only, subscription model
Stevie II	Trinity College Dublin	N/A	Social robot, care home	Institutional setting, not home deployment

2.3). Fall Detection Methods

Wearable accelerometers provide accurate fall detection, with reported sensitivity and specificity exceeding 95%, but their efficacy depends on continuous user compliance [23]. Vision-based fall detection systems monitor human posture and shape deformation, achieving detection rates around 92%, but raise privacy concerns and struggle with occlusions [24]. Floor sensor networks can detect falls unobtrusively without requiring wearable devices, although installation costs and spatial limitations remain challenging [25]. AI-based fall detection using computer vision and pose estimation allows real-time detection with standard RGB cameras, but performance can degrade under poor lighting or occlusions (Liu et al., 2020).

Vision-based fall detection systems that use pose estimation frameworks such as MediaPipe have been shown to detect falls with high accuracy (~95.8%), extracting key points from standard video streams without the need for additional sensors, though challenges remain in real-world lighting and camera conditions [26]. Systematic reviews of fall detection systems indicate that multiple sensing modalities — including wearable sensors, vision-based approaches, and hybrid solutions — have been developed for elderly monitoring, with varying performance characteristics depending on the sensor type and algorithm used [27].

2.4). Robot Navigation and Indoor Mobility

A SLAM-based navigation system was demonstrated on the Pepper platform, achieving reliable indoor localization and path planning using ROS, and maintained low mean localization errors while interacting with dynamic environments [28]. Autonomous navigation systems built on ROS can integrate SLAM with path planning (e.g., A* and DWA) to create accurate maps and allow dynamic path planning in indoor environments [29].

Comparative evaluations of ROS SLAM modules such as GMapping and RTAB-Map reveal differences in mapping quality and navigation performance, underscoring the importance of algorithm selection for indoor robot mobility [30]. Recent reviews emphasize semantic navigation in indoor environments, where robots enrich geometric maps with contextual information to improve localization, obstacle avoidance, and interaction with humans [31]. Hybrid motion planning frameworks such as RL-DWA integrate reinforcement learning with local planners to navigate toward and follow users in cluttered indoor spaces, showcasing advanced human-robot mobility strategies [32].

2.5). GSM and IoT-Based Health Alerting

IoT-based health monitoring systems that combine vital sign sensing (e.g., heart rate, SpO_2, temperature), fall detection, GPS tracking, and GSM alerting can deliver real-time SMS notifications to caregivers, enabling prompt intervention without requiring Wi-Fi or constant internet connectivity [33]. Real-time health monitoring architectures that deploy GSM modules for data transmission allow vital signs like temperature and pulse to be sent to caregivers through SMS, making them suitable for environments with limited internet access [34].

Remote health monitoring frameworks can integrate multisensory biomedical data with fall detection and GSM SMS alerting mechanisms, allowing caregivers to be notified of abnormalities even when a Wi-Fi network is unavailable [35]. Simple remote monitoring systems using accelerometers and physiological sensors can be designed to automatically send GSM SMS alerts when abnormal patterns or falls are detected, even in offline scenarios [36]. Systematic surveys of IoT-based fall detection highlight the integration of sensor networks and communication modules (including GSM and cellular) to deliver timely alerts, emphasizing practical considerations for real-world deployment [37].s

2.6). Human-Robot Interaction with Elderly Users

Human-robot interaction plays a critical role in enabling assistive robots to support independent living for older adults. Studies show that perceived usefulness, ease of use, safety, reliability, and emotional connection strongly influence the acceptance of robotic assistants [38]. Studies evaluating social robots deployed in real home environments show frequent daily interaction by elderly users and indicate that robots providing practical assistance and natural communication are more readily accepted [39].

Research indicates that several psychological and usability factors influence older adults' interaction with robots, including privacy concerns, unfamiliarity with technology, trust issues, and the "uncanny valley" effect associated with highly human-like robots [40]. Socially assistive robots designed for elderly care should support natural communication, context-aware interaction, and integration with smart environments to effectively assist daily activities [41]. Non-verbal communication cues such as gestures, touch interaction, and visual feedback can significantly improve trust and engagement in human-robot interaction with elderly users [42].

2.7). Robotics in Developing Countries

Rehabilitation robots have shown the potential to improve health outcomes, but their high costs and resource requirements limit availability in low- and middle-income countries. Strategies such as reducing mechanical complexity and using affordable materials have been proposed to make rehabilitation robotics more accessible in these settings [43]. Research on eldercare robotics in emerging economies highlights systemic barriers such as unequal distribution of technology, limited access to resources, and policy gaps, which continue to restrict the benefits of robotic care for older adults with disabilities [44].

Robotics technologies have the potential to support sustainable healthcare and reduce inequality, yet applications tailored to low-resource settings, including developing countries, have been underexplored in the literature [45]. Studies within Latin America show that social and assistive robots are rarely adopted in healthcare due to high costs, indicating a need for open-source and low-cost platforms adapted to local contexts [46]. Humanitarian robotics initiatives such as WeRobotics demonstrate community-centered efforts to leverage robotics for local needs, though such approaches remain limited compared with mainstream technology development [47].

2.8). Medication Adherence Technology

IoT-based smart pillbox systems can generate real-time reminders and alert caregivers if medications are missed, improving adherence in elderly patients [48]. AI-enhanced medication adherence systems that deliver adaptive reminders via mobile applications can increase compliance among older adults by personalizing alerts and monitoring behavior [49].

RFID-assisted platforms that detect medication removal from pillboxes and confirm ingestion can provide a reliable method of adherence monitoring, with alerts sent if doses are missed [50]. Smart medication dispensers equipped with embedded sensors and communication modules can track pill intake and send reminders or alerts to caregivers when medication is missed [51]. Wearable IoT medication monitoring frameworks enhance adherence by detecting patient activities and issuing adaptive reminders, particularly aiding those with mobility limitations [52]. Recent systematic reviews demonstrate that mobile health interventions, including smartphone apps with reminders and interactive features, can significantly improve medication adherence in patients with chronic conditions, though integration with health systems and usability remain important considerations [53].

2.9). Research Gap and Positioning

Table 2 Research Gap — Existing Solutions vs CARA

Dimension	PARO	PEPPER	ElliQ	Wearables	Smart Home	CARA
Cost	$5,000+	$15,000+	$250/month	$100–500	$200–1,000	<$500 one-time
Wi-Fi Required	Yes	Yes	Yes	No	Yes	No — GSM only
Fall Detection	No	No	No	Yes	No	Yes — MediaPipe AI
Developing World Design	No	No	No	Partial	No	Yes — explicit target
Elderly User Interaction Required	Minimal	Yes	Yes	Yes (button)	Yes	Zero — fully passive
Companion Features	Basic	Advanced	Advanced	None	None	15 features

Analysis of current assistive technologies for elderly care reveals significant variation in cost, connectivity requirements, functional capabilities, and suitability for deployment in low-resource environments. High-end social robots such as PARO and PEPPER provide advanced companion features and interaction but are expensive and require Wi-Fi connectivity, limiting accessibility in developing contexts. Solutions like ElliQ offer subscription-based services with companion functionalities but similarly depend on continuous internet access. Wearables and smart home systems provide partial solutions for monitoring and fall detection, with lower cost and intermittent interaction requirements; however, their functionality is often task-specific and lacks full integration for comprehensive eldercare. Notably, CARA emerges as a unique platform explicitly designed for developing-world settings, incorporating low-cost, GSM-compatible monitoring, real-time fall detection using MediaPipe AI, and multiple passive companion features without requiring active user interaction. These comparisons highlight the need for integrated, affordable, and connectivity-flexible assistive systems capable of supporting diverse elderly populations in both high- and low-resource contexts.

2.10). Chapter Summary

This chapter reviewed the current state of elderly care and assistive robotics, highlighting demographic trends, technological solutions, and persistent challenges. Sri Lanka, like many developing countries, is experiencing rapid population aging, creating increasing demand for long-term care and support systems. Existing assistive robots—including PARO, Pepper, Care-O-Bot, and ElliQ—demonstrate benefits in emotional support, social engagement, and task assistance; however, high costs, dependence on Wi-Fi, and limited suitability for low-resource environments constrain their broader adoption.

Fall detection methods, ranging from wearable sensors to vision-based AI systems such as MediaPipe, provide critical safety monitoring, yet each modality has trade-offs in accuracy, privacy, and environmental requirements. Robot navigation research emphasizes SLAM-based systems and hybrid motion planning strategies, demonstrating reliable indoor mobility and human-following capabilities, though real-world deployment remains limited. GSM and IoT-based health alerting systems offer practical solutions for remote monitoring in connectivity-limited settings, reinforcing the importance of low-infrastructure designs. Human-robot interaction studies reveal that elderly acceptance depends on usability, trust, emotional connection, and seamless integration into daily routines. Research on robotics in developing countries highlights systemic barriers—including affordability, resource access, and policy limitations, underscoring the need for context-sensitive designs. Medication adherence technologies, spanning IoT pillboxes, AI-enabled mobile reminders, RFID confirmation, and wearable frameworks, illustrate the potential to improve compliance while revealing usability and integration challenges.

The synthesis of these studies identifies a clear gap: existing solutions often address individual aspects of eldercare but rarely integrate affordability, accessibility, fall detection, autonomous monitoring, and user-centered interaction within a single, low-cost platform suitable for developing world contexts. CARA is positioned to address this gap, combining GSM-based communication, AI-powered fall detection, passive interaction, and companion features, thereby advancing the field toward comprehensive, practical, and inclusive elderly care solutions.

3). Chapter: System Design

3.1). Design Requirements

3.1.1). Functional Requirements

Table 3 List of Functional Requirements (Author Developed)

Req. ID	Requirement	Source	Priority
FR1	The system shall detect fall posture using computer vision with minimum 80% accuracy across 20 standardized trials	Objective 2	Critical
FR2	The system shall deliver SMS alerts to registered family members within 30 seconds of fall confirmation, without Wi-Fi	Objective 3	Critical
FR3	The system shall avoid obstacles during navigation, stopping at minimum 20cm from any obstacle	Objective 1	Critical
FR4	The system shall trigger medication reminders at 4 programmed daily times with RFID confirmation	Objective 4	High
FR5	The system shall send SMS alerts to family if medication is not confirmed within 15 minutes of reminder	Objective 4	High
FR6	The system shall allow family members to view live camera feed via phone browser	User need	Medium
FR7	The system shall allow family members to drive CARA remotely via phone browser	User need	Medium
FR8	The system shall display warm greeting messages on LCD when person is detected	User need	Medium
FR9	The system shall provide LED visual feedback indicating current system state	User need	Low
FR10	The system shall log key events (fall detection, navigation, medication reminders) for offline analysis of system behavior and evaluation.	Objective 5	Low

3.1.2). Non-Functional Requirements

Table 4 List of Non-Functional Requirements (Author Developed)

Req. ID	Requirement	Metric	Priority
NFR1	Total component cost shall not exceed $500 USD	Component receipts	Critical
NFR2	Core safety features shall operate without Wi-Fi using GSM only	Test without Wi-Fi	Critical
NFR3	Elderly users shall not be required to interact with CARA for core functions	Zero-interaction test	Critical
NFR4	System shall operate on battery for minimum 90 minutes under full load	Battery life test	High

3.2). System Architecture Overview

Table 5 System Architecture Short Overview (Author developed)

```
Camera → Jetson Nano → MediaPipe Fall Detection → ROS Topic
                                ↓
                          Serial Bridge
                                ↓
                        Arduino Controller
                                ↓
        Motors | GSM | RFID | LCD | PIR | Ultrasonic
```

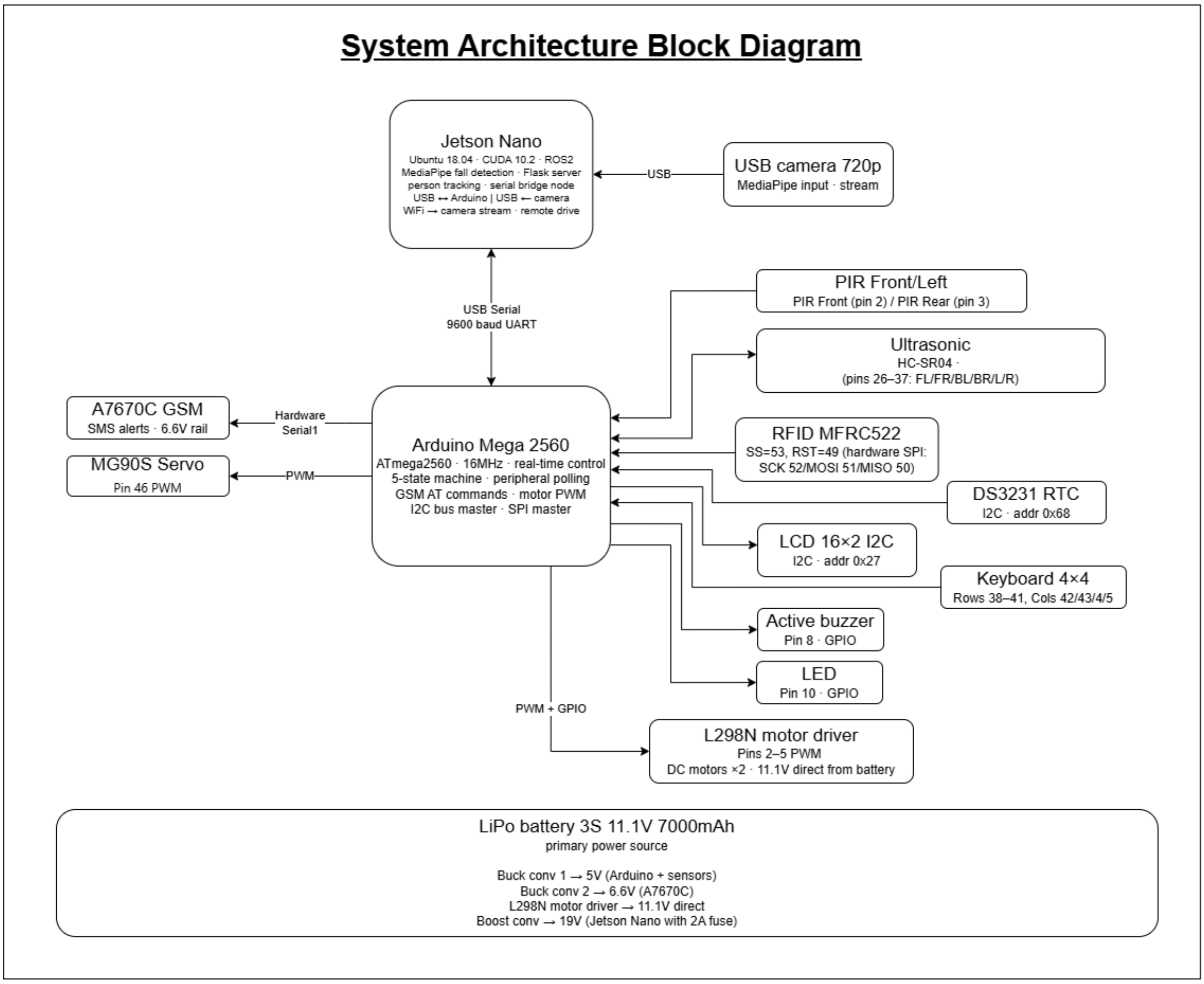

Figure 3-1 System Architecture Block Diagram (Author developed)

3.3). Software Architecture

Table 6 Arduino State Machine — All 5 States and ROS Nodes (Author Developed)

Component	Platform	State / Node Name	Trigger	Action
State 1	Arduino	IDLE	System start / reset	LCD: CARA Ready · LED: green slow pulse · PIR polling every 100ms · motors off · MediaPipe: OFF (saves GPU) person_lock: inactive · all ultrasonic sensors polling
State 2	Arduino	ACTIVE	PIR detects person	MediaPipe: ON · person_lock activates color band HSV filter · Layer 1 RFID UID confirms target identity · Layer 2 orange band HSV tracks target continuously · nose

Component	Platform	State / Node Name	Trigger	Action
				X vs frame center → STEER_LEFT or STEER_RIGHT commands via serial · upper ultrasonic ring × 6 controls approach speed · motors approach at 0.3m/s · pan-tilt head turns toward person · LED: amber pulse · LCD: Person detected · if TARGET_LOST → return to IDLE
State 3	Arduino	MONITORING	Ultrasonic < 60cm	Motors: STOP · MediaPipe: fall detection active · upper ultrasonic ring × 6 monitors distance continuously · if dist. >80cm → return to ACTIVE · LED: amber steady · LCD: Watching over you · pan-tilt: locked on person · await FALL_DETECTED serial command from Jetson Nano
State 4	Arduino	ALERT	Receives 'FALL_DETECTED'	Buzzer: ON continuous · LCD: Help is coming! · LED: red fast blink · sendSMS() to registered family number · motors: locked STOP · camera stays on person · pan-tilt: locked · 30s timer starts · obstacle override still active
State 5	Arduino	RESET	30s timer expires	Buzzer: OFF · LED: green · LCD: CARA Ready · all sensors reinitialized · MediaPipe: restart · person_lock: reset · automatic transition → STATE 1 IDLE
ROS Node	Jetson Nano	fall_detection	Camera frame received	MediaPipe pose estimation at 12–15fps · check LEFT_HIP y>0.77 · RIGHT_HIP y>0.77 · LEFT_SHOULDER y>0.55 · all conditions true for 2s → fall confirmed · publish /fall_detected topic · also outputs nose X position to pan_tilt node for steering
ROS Node	Jetson Nano	serial_bridge	fall_detected topic	Subscribes /fall_detected and /target_visible topics · sends full command stream to Arduino via USB serial at 9600 baud: FALL_DETECTED · STEER_LEFT · STEER_RIGHT · MOVE_FORWARD · SLOW_DOWN · STOP · TARGET_LOST · MOVE_BACKWARD · TURN_LEFT ·

Component	Platform	State / Node Name	Trigger	Action
				TURN_RIGHT · PAN_LEFT · PAN_RIGHT
ROS Node	Jetson Nano	camera_stream	HTTP request from browser	Flask MJPEG server · port 5000 · /video_feed endpoint · runs as daemon thread · streams from /dev/video0 · accessible from phone browser on same Wi-Fi · does not interfere with fall_detection (separate thread)
ROS Node	Jetson Nano	person_lock	Camera frame received	OpenCV HSV filter detects orange band · Layer 1 RFID UID registers target identity · Layer 2 color band tracks continuously · band absent → publish TARGET_LOST · others in room ignored · publishes /target_visible topic
ROS Node	Jetson Nano	pan_tilt	Nose X position from fall_detection + PIR direction signal	Calculates pan servo angle from nose X vs frame center 320px · PIR direction → initial head turn · default tilt 75° · sends PAN_LEFT / PAN_RIGHT via serial_bridge · Pin 9 pan · Pin 10 tilt
ROS Node	Jetson Nano	remote_control	HTTP POST from phone browser	Receives MOVE_FORWARD / MOVE_BACKWARD / TURN_LEFT / TURN_RIGHT / STOP · forwards via serial to Arduino · obstacle override cannot be disabled even during remote drive
ROS Node	Jetson Nano	Health management	DS3231 RTC time match	Triggers medication reminder × 4/day at 08:00 13:00 18:00 22:00 · buzzer + LCD: Medicine Time! · RFID poll → confirm · 15min no confirm → sendSMS() missed medication alert · 12:00pm daily check-in LCD reminder · runs alongside all 5 main states

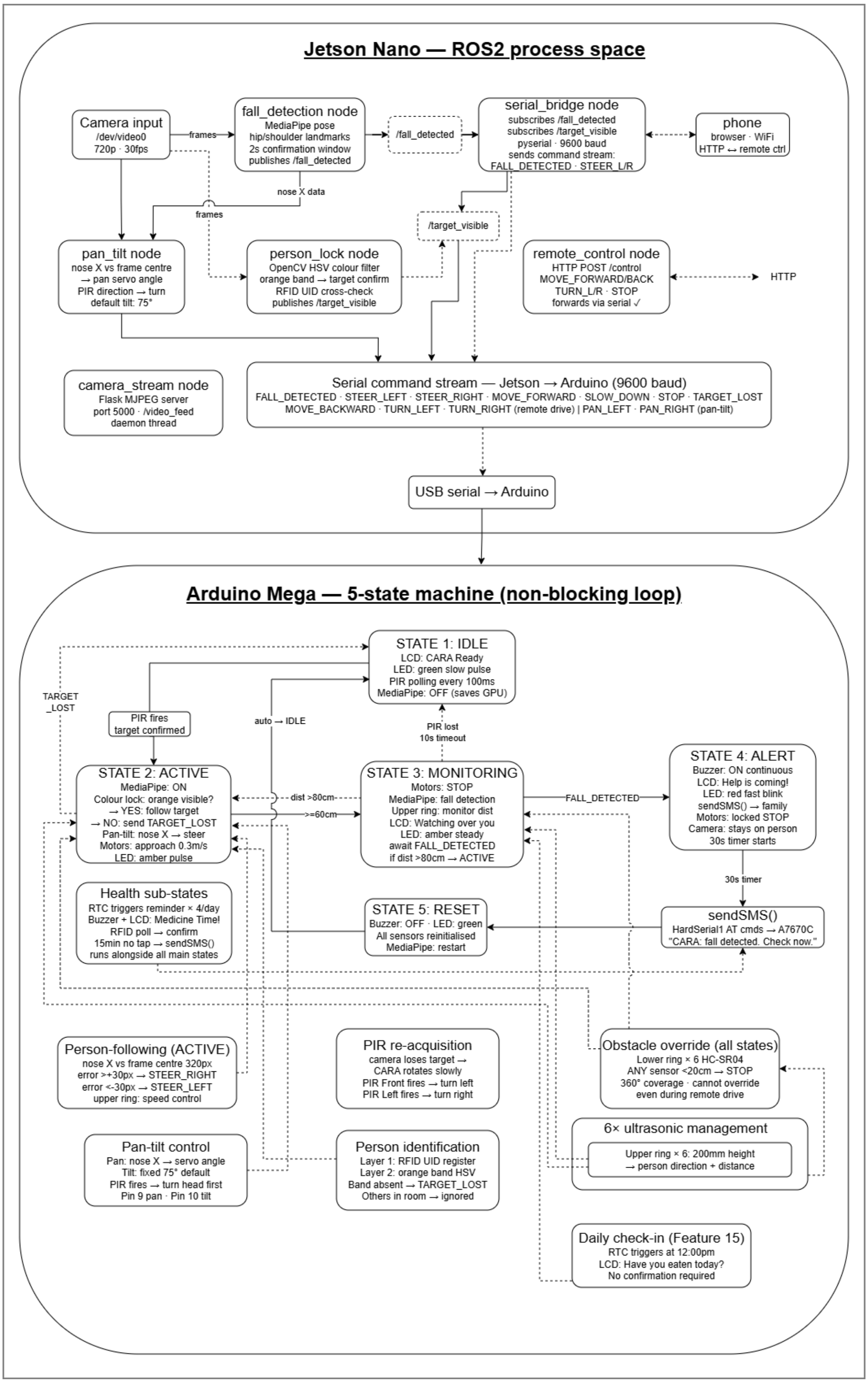

Figure 3-2 Software Architecture with ROS nodes and Arduino state machine (Author Developed)

3.4). Hardware Architecture

Table 7 Hardware Component Specification (Author Developed)

Component	Model	Specification	Purpose	Cost (USD)	Justification
AI Processor	NVIDIA Jetson Nano 4GB	4-core ARM, 128 CUDA cores, Ubuntu 18.04	MediaPipe fall detection, camera streaming, ROS	$99–150	Only affordable SBC with GPU for MediaPipe at 15fps
Microcontroller	Arduino Mega 2560	ATmega2560, 54 digital I/O, 16 analog pins, 256KB flash, 16MHz	Peripheral control, sensor polling, GSM, state machine	$10–25	Widely supported, sufficient I/O, real-time reliability
Camera	USB Webcam 720p	720p 30fps, USB 2.0, UVC compatible	MediaPipe pose input, family camera feed, Visual SLAM input, fall detection	$15–30	UVC compatibility ensures Jetson Nano driver-free operation
Fall Detection	MediaPipe Pose (ML model)	33 body landmarks, 15fps on Jetson Nano	Real-time posture analysis for fall detection	Free (open source)	Runs on Jetson Nano without GPU server requirement
PIR Sensor	HC-SR501	3–7m range, adjustable sensitivity	Person presence detection	$1–3 each	Low power, simple digital output, proven range
Ultrasonic	HC-SR04	2–400cm range, 15° beam angle	Obstacle detection and SLAM correction	$2–5	Industry standard for short-range distance measurement
GSM Module	A7670C LTE module	Quad-band GSM, AT command interface	SMS emergency alerts — Wi-Fi-free communication	$5–12	LTE Cat-1 coverage in regions where 2G has been discontinued

Component	Model	Specification	Purpose	Cost (USD)	Justification
Motor Driver	L298N Dual H-Bridge	1.2A per channel, PWM speed control	DC motor direction and speed control	$3–8	Dual-channel in single IC reduces board space
DC Motors x2	6V Gear Motor, 150RPM	6V, 150RPM, 65mm wheel compatible	Robot mobility	$5–10 each	Adequate torque for 2kg robot at 0.3m/s on flat surface
RFID Reader	MFRC522	13.56MHz, SPI interface, 3–5cm read range	Medication confirmation	$2–5	Standard for contactless identification, short-range intentional
RTC Module	DS3231	I2C, ±2ppm accuracy, battery backup	Medication reminder timing without network sync	$2–5	Battery-backed maintains time through power cycles
LCD Display	16x2 with I2C backpack	16 char × 2 lines, I2C at 0x27	State messages and greeting display	$2–5	I2C reduces pin usage to 2 wires for both LCD and RTC
Servos	MG90S 9g Micro (metal gear)	180° rotation, ~2.2kg/cm torque	Greeting wave gesture	$3–6 each	Metal gears provide the extra torque the SG90 couldn't for the wave gesture
LiPo Battery	3S 11.1V 7000mAh	11.1V nominal, 30C discharge, T-connector	Primary power for all systems	$15–30	Energy density and discharge rate appropriate for 90min target
Buck Converters	LM2596 Step-Down	Input 4–40V, Output 1.25–37V, 3A	5V for Arduino + Sensors; 6.6V regulated for A7670C module	$1–3 each	A7670C requires regulated 6.6V — cannot use Arduino 5V pin
Boost Converter	XL6009	Input 3–32V, Output 5–35V	19V for Jetson Nano	$1-3	Steps 11.1V up to the Jetson's required 19V input at low cost, with headroom for the 2A fuse

3.5). Physical Design

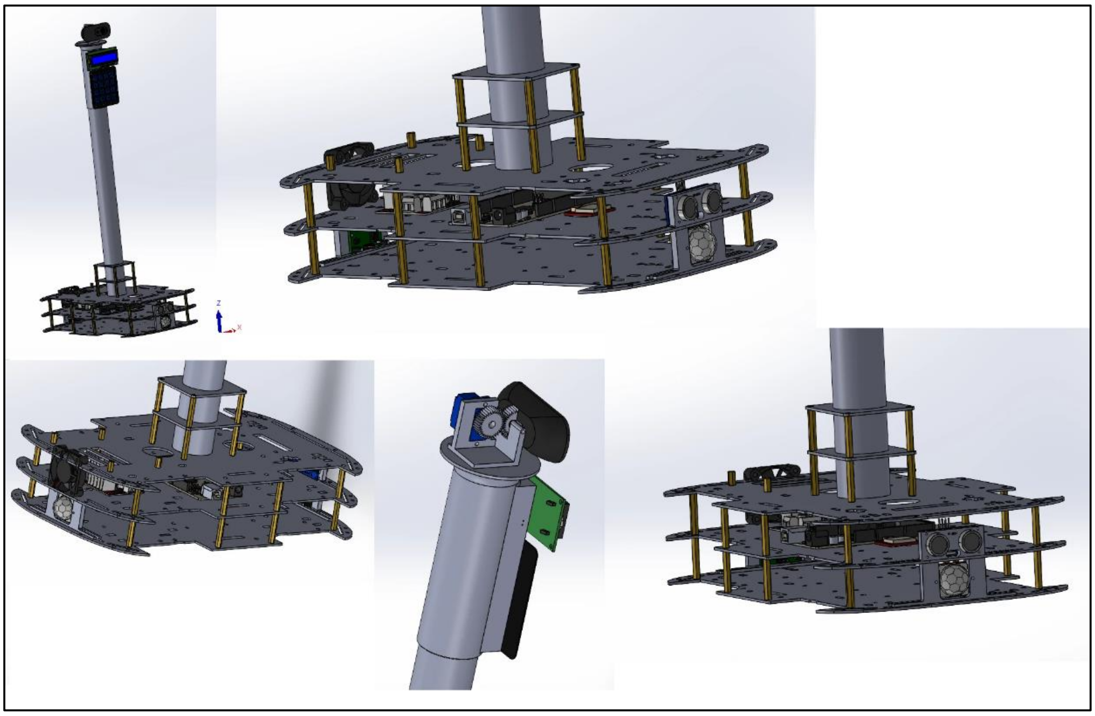

Figure 3-3 Physical Design using SolidWorks (Source: Author developed)

3.6). Power System Design

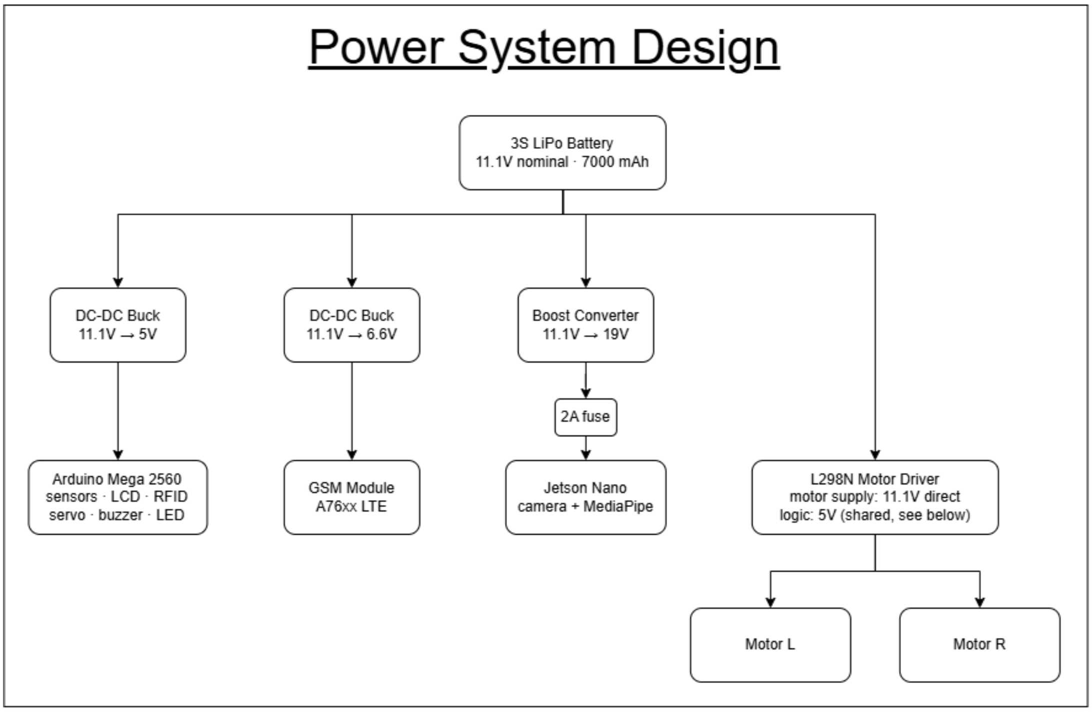

Figure 3-4 Power System Design (Source: Author developed)

CARA is powered by a single 3-cell (3S) LiPo battery rated at 11.1V nominal and 7000 mAh, sized to meet the requirement of at least 90 minutes of continuous operation under full load. Four independent rails are derived from this one source. A dedicated DC-DC buck converter steps the battery down to a regulated 5V rail for the Arduino Mega 2560 and its attached sensor and peripheral bus (the ultrasonic ring, PIR sensors, RFID reader, LCD, servo, and buzzer). A second buck converter regulates the battery down to 6.6V for the GSM module, kept on its own rail so the module's high-current transmit bursts cannot brown out the microcontroller. A third DC-DC converter supplies the Jetson Nano at a regulated 19V, rated for approximately 2A to cover its peak draw under simultaneous camera capture and MediaPipe inference. The L298N motor driver is powered directly from the unregulated 11.1V battery rail, since the drive motors require more voltage and current than any regulated rail provides; the driver's own logic pins share the same regulated 5V rail as the Arduino Mega, keeping all logic-level grounds common across the system.

This mirrors the power-domain-separation principle used elsewhere in CARA's design: high-current or noise-generating loads — the GSM module during transmission, the motors under load — are each isolated on their own supply, separate from the lower-power logic and sensor rail, to minimize sbrownout risk and noise coupling into the microcontroller.

3.7). Communication Design

CARA integrates three distinct communication protocols, each chosen to satisfy the specific functional and data requirements of its subsystems. The architecture includes UART serial communication between Jetson Nano and Arduino, I2C communication between Arduino and peripheral modules (LCD and RTC), and GSM-based communication via AT commands between Arduino and the A7670C module. Among these, GSM serves as the primary external communication channel to ensure reliable message delivery in rural and underdeveloped regions where internet infrastructure may be limited or unavailable. Wi-Fi is intentionally excluded from all safety-critical communication paths as a deliberate system design decision, ensuring independence from local network availability rather than reflecting a system constraint.

UART Serial Communication (Jetson Nano ↔ Arduino)

UART serial communication is used to transmit fall detection events from the Jetson Nano to the Arduino. The system operates at a baud rate of 9600 bps, which provides sufficient reliability and stability for short command-based messages such as *FALL_DETECTED*. The Jetson Nano sends data as a string, which the Arduino receives using *Serial.readStringUntil('\n')*. This communication is designed as a one-way signaling mechanism where the Jetson acts as the decision-making unit and the Arduino executes physical responses such as activating a buzzer or updating the LCD display. Explicit acknowledgements are not required, as system feedback is provided through these physical indicators. On the Jetson side, the Arduino typically appears as a USB serial device such as */dev/ttyUSB0* or */dev/ttyACM0*.

I2C Bus Communication (LCD + RTC)

The I2C bus is used to connect the LCD display (address 0x27) and the DS3231 real-time clock module (address 0x68) to the Arduino over a shared two-wire interface. This protocol enables multiple devices to operate on the same bus using unique addressing, thereby reducing the number of required GPIO pins and improving hardware efficiency. The shared lines consist of SDA and SCL, typically connected to Arduino pins A4 and A5 respectively. Each peripheral device is individually addressed by the master controller, ensuring conflict-free communication while maintaining scalability for additional I2C devices if required.

GSM Communication via AT Commands (A7670C)

GSM communication is implemented using the A7670C module to send SMS alerts in response to critical events such as fall detection or missed medication reminders. This represents the most important communication pathway in CARA, enabling direct user notification independent of local internet connectivity. The module is controlled using Hardware-Serial1 (commonly on Arduino pins 18 and 19) and communicates via the standard Hayes AT command set. Key commands include **AT** (module test), **AT+CSQ** (signal quality check), **AT+CMGF=1** (SMS text mode configuration), and **AT+CMGS** (message transmission).

This GSM-based approach ensures system operability in rural and infrastructure-limited regions, directly addressing the deployment constraints identified in Chapter 2. The A7670C operates with 6.6V supply range and therefore requires a dedicated regulated power supply (e.g., a buck converter) separate from the Arduino 5V rail to ensure stable operation.

Overall, the selected communication protocols are optimized to balance reliability, hardware efficiency, and operational independence from local network infrastructure. In particular, the GSM communication layer enables CARA to function effectively in real-world developing-world scenarios, aligning the system design with the intended research objectives and deployment environment.

3.8). Safety Design

Safety is a fundamental design requirement for robotic systems intended to operate in close proximity to elderly or vulnerable users. In CARA, safety has been treated as a primary system constraint rather than an auxiliary feature. The overall design is informed by ISO 13482:2014, the international safety standard for personal care robots. Although full certification is beyond the scope of this prototype implementation, the standard provides a structured reference framework for evaluating design decisions and risk mitigation strategies.

Physical Collision Safety

CARA employs an HC-SR04 ultrasonic sensor for continuous real-time distance monitoring during navigation. When an obstacle is detected within a threshold distance of 20 cm, the system immediately halts motor operation regardless of the current control state or command source. This threshold is selected based on the robot's maximum operating speed, estimated stopping distance, and motor response latency to ensure safe deceleration under worst-case conditions.

The emergency stop mechanism is designed to be non-overridable. Even in cases where remote navigation commands are issued via the web interface, obstacle detection retains absolute priority. This design directly aligns with ISO 13482 requirements that personal care robots must avoid causing harm through unintended physical interaction.

Fall Detection and Alert Safety

To improve robustness against false triggers, the fall detection module includes a 2-second confirmation window before a fall event is officially registered. This temporal filtering reduces the likelihood of single-frame misclassifications generating false alarms. Additionally, a 30-second cooldown period is enforced after each alert transmission to prevent repeated SMS notifications being sent to caregivers in rapid succession.

The system targets a false positive rate of fewer than two alerts per hour under continuous operation, with planned validation through extended real-time testing sessions.

Electrical and Power Safety

Electrical safety is ensured through careful power domain separation and regulation. The A7670C GSM module is powered using a regulated 6.6V supply via a dedicated buck converter, as it cannot be reliably powered from the Arduino 5 V rail due to high current demand during transmission bursts. A 1000 µF capacitor is placed across the A7670C power terminals to stabilize transient current spikes that occur during SMS transmission and to prevent system resets.

The LiPo battery system includes an integrated protection circuit to prevent over-discharge, thereby reducing the risk of battery degradation and thermal hazards. In addition, high-current subsystems such as motors and GSM communication are electrically isolated from low-power sensor and logic circuits through separate power rails to minimize noise coupling and ensure system stability.

Operational Safety Limitations

CARA v1.0 is implemented as a research prototype and has not undergone formal certification under ISO 13482 or equivalent regulatory frameworks. Any real-world deployment in eldercare environments would require additional validation, including extended user trials, clinical safety assessments, and compliance verification.

These limitations are acknowledged as part of the system scope and are addressed further in the future work outlined in Chapter 6.

Safety in CARA is implemented across three layers: physical safety through obstacle avoidance, functional safety through validated fall detection logic, and electrical safety through regulated power design. Collectively, these mechanisms work together to reduce operational risk and enhance system reliability for both user interaction and internal hardware stability.

4). Chapter : Implementation

4.1). Introduction

This chapter describes the implementation process for CARA Version 1.0, from initial hardware assembly through full system integration. The implementation followed the system design specifications detailed in Chapter 3, with the objective of realizing all 15 functional features within the established cost constraints.

The implementation proceeded through five phases: hardware assembly, Arduino firmware development, Jetson Nano AI integration, GSM communication setup, and full system integration. Each phase presented unique challenges that required iterative problem-solving. This chapter documents the technical solutions applied and the rationale behind design decisions.

4.2). Hardware Implementation

The hardware is organized into four functional subsystems, all interfaced through a single Arduino Mega 2560.

Sensing. Six HC-SR04 ultrasonic sensors form a complete obstacle-detection ring around the chassis (front-left, front-right, back-left, back-right, left, right), each using a dedicated trigger/echo pin pair. Two PIR motion sensors (front and rear) serve as the primary presence-detection trigger that wakes the robot from IDLE.

Identification and input. An MFRC522 RFID reader is connected over the Mega's hardware SPI bus for medication-confirmation tap events. A 4×4 matrix membrane keypad provides manual input, with # mapped to a panic/alert trigger and * to acknowledge and silence an active alert.

Timekeeping and display. A DS3231 real-time clock and a 16×2 I2C LCD were originally connected to a shared hardware I2C bus, on the basis that both peripherals are low-bandwidth and infrequently polled. During implementation, one clock-line pin on the assembled Mega was found to be physically damaged, confirmed by direct voltage-level testing independent of any library code and corroborated by both modules working correctly when tested on a separate microcontroller. The LCD was recovered by re-implementing the I2C protocol in software on a different, undamaged pin pair; the real-time clock remains disabled pending an equivalent fix, since its communication protocol is less straightforward to reimplement at that level (see Limitations, Chapter 5).

Actuation and feedback. Robot movement is driven through an L298N dual H-bridge motor driver, controlled via four direction pins and two PWM speed pins. A buzzer provides audible alerts, and a PWM-driven LED provides an ambient "breathing" status indicator during IDLE.

An MG90s servo performs the greeting wave gesture; it is deliberately wired to its own timer group, separate from the motor PWM pins, to prevent servo attach/detach operations from causing glitches in motor control signals.

Communication. A GSM A7670C module is wired to the Mega's hardware Serial1 port for SMS alerting. A USB serial connection to the Jetson Nano/laptop, running at 9600 baud, carries the command protocol used for fall alerts (FALL_DETECTED) and remote-drive commands (MOVE_FORWARD, TURN_LEFT, etc.).

4.3). Arduino Firmware

The Arduino Mega firmware is organized as a five-state finite state machine — IDLE, ACTIVE, MONITORING, ALERT, and RESET_STATE — implemented entirely on Millis()-based non-blocking timing, with no delay() calls inside the state logic itself. This allows the main loop to continue servicing serial commands, the keypad, RFID reads on every cycle regardless of which state the robot is in.

In IDLE, the robot pulses its status LED using a sine-wave brightness curve and monitors two PIR sensors (front and rear) for motion. When either sensor triggers, the robot displays a randomly selected greeting on its LCD, performs a servo wave gesture, and transitions to ACTIVE. In ACTIVE, the robot drives forward autonomously toward the detected person, but this movement is gated by the front-left and front-right ultrasonic sensors; if an obstacle is detected within the stop threshold, the robot halts and transitions to MONITORING. If no PIR activity is registered for three seconds, the robot assumes the person has left and returns to IDLE. In MONITORING, the robot holds position and signals the Jetson Nano over serial (START_FALL_DETECTION) to begin camera-based pose analysis. This state times out to IDLE after three seconds of inactivity. ALERT can be entered from two independent sources — a FALL_DETECTED command received from the Jetson, or a manual panic press on the keypad's # key — demonstrating that the state machine decouples the source of an alert from its response. During ALERT, a non-blocking buzzer pattern sounds and an SMS alert is issued (currently logged to serial pending final GSM module integration); the alert clears automatically after thirty seconds, or early if acknowledged via the keypad's * key.

A remoteControl flag provides a clean override mechanism for the browser-based manual drive interface: when a human is actively driving the robot, this flag suppresses the autonomous forward-approach logic in ACTIVE, while the underlying obstacle-avoidance checks remain active regardless of drive mode — ensuring manual control cannot bypass collision safety.

4.4). Jetson Nano and Fall Detection

Fall detection runs on a two-layer confirmation pipeline designed to reject single-frame noise while keeping detection latency low. Every camera frame is passed through MediaPipe's Pose Landmarker (lite model) to extract 33 body landmarks, from which 69 features are derived — normalised x/y coordinates relative to the hip midpoint and body height, plus body angle, body aspect ratio, and mean landmark visibility. These features are passed to a trained RandomForestClassifier (200 estimators, class-balanced) that outputs a per-frame fall-confidence score.

In Layer 1, any frame scoring at or above a 0.35 confidence threshold enters a tracking state. In Layer 2, while tracking, the system re-checks body angle and hip position every 0.3 seconds; three consecutive checks showing a rising angle or rising hip position confirm a fall, giving a total confirmation window of approximately 2.7 seconds from initial trigger to alert. This window was deliberately kept short to minimise the delay between an actual fall and the SMS alert reaching family members, while the three-check requirement still filters out momentary pose misreads. If confidence rises but the pose does not continue falling within this window, the system instead classifies the event as a static pose (e.g. sitting or lying down deliberately) rather than a fall. A 30-second cooldown prevents repeated alerts for the same event.

The classifier was trained on 644 labelled pose samples (303 fall, 341 not-fall) and evaluated with an 80/20 train-test split plus 5-fold cross-validation. It achieved 87.6% test accuracy and 89.6% cross-validated accuracy (±4.8%), with 84.6% precision and 90.2% recall on the fall class (confusion matrix: 55 true positives, 6 false negatives, 10 false positives, 58 true negatives). The 90.2% recall was prioritised over precision by design, since missing a real fall is a more serious failure than an occasional false alert for this application. On a confirmed fall, the Jetson sends a FALL_DETECTED command to the Arduino over USB serial at 9600 baud, which triggers the ALERT state described in the Arduino Firmware section.

4.5). GSM Integration

GSM alerting was first developed around a SIM800L module, communicating over AT commands via the Arduino Mega's hardware Serial1 port. The AT-command protocol layer was fully validated in standalone testing: the module correctly handshakes, accepts echo-off configuration, reports supply voltage, and accepts the text-mode SMS command sequence (AT+CMGF=1, AT+CMGS, Ctrl+Z-terminated message body).

Network registration, however, could not be achieved, and this was investigated as a staged elimination of possible causes. A SIM-specific restriction was tested first by swapping between two different SIM cards; both failed to register in the same way, ruling this out. Supply power was investigated next, since the SIM800L's transmit current bursts are a well-documented cause of brownout on inadequately specified supplies; increased supply voltage was tested, and a stabilising capacitor across the supply rails was identified as the standard hardware fix for this failure mode, but registration still could not be confirmed. The root cause was ultimately identified as network-level incompatibility rather than a wiring, power, or SIM issue: the

SIM800L is a 2G/GPRS-only module, and UAE mobile carriers have fully discontinued 2G service, making network registration structurally impossible on this module regardless of any wiring or power correction.

This finding informed a decision to migrate to a SIM7600-series module, which supports UAE's active LTE bands. Because the AT-command interface for SMS sending is consistent across the SIM800/SIM7s600 family, the command sequence already developed and validated at the protocol level remains directly applicable once the new module is integrated. Within the main firmware, GSM logic remains present as a commented block inside sendSMS(), with a serial-print stub standing in until the SIM7600 module is procured and integrated.Within the main firmware, GSM logic remains present as a commented block inside sendSMS(), with a serial-print stub standing in until the SIM7600 module is available and integrated.

Following the network-incompatibility finding above, an A76xx-series LTE Cat-1 module was procured as a direct replacement. Bring-up required addressing several further issues not present with the earlier module: the replacement module's main UART interface operates at 1.8V logic rather than 5V; its official current draw (5–8.4V input, several hundred mA under transmit) required a dedicated regulated supply rather than the microcontroller's own rail; and initial communication exhibited intermittent single-character corruption at its fixed 115200 baud rate, resolved by migrating from software-emulated serial to the Arduino Mega's dedicated hardware serial port and loosening string-matching in the response parser to tolerate occasional corrupted bytes. Once stable, a complete interactive SMS system was implemented and validated: the module correctly sends outbound messages, receives and confirms delivery reports, and logs inbound messages together with sender number and timestamp, fulfilling this objective on the replacement hardware.

4.6). Phase 1 System Integration

Phase 1 integration refers to the point at which every sensor and actuator described above — previously developed and validated as isolated test sketches under Sensor_Testing_Codes/ — was consolidated into a single coordinated firmware (cara_main_mega.ino) governed by one central state machine, rather than operating independently.

Integration was verified through a built-in startup self-test routine, executed automatically on boot, which exercises the status LED, buzzer, the full six-sensor ultrasonic ring, and both PIR sensors, reporting results to both the LCD and the serial console before the system enters IDLE. This routine serves as the primary integration-verification mechanism for the assembled hardware.

Several cross-subsystem interactions were demonstrated working as a result of this integration: PIR-triggered presence detection leading to autonomous forward movement gated by real-time obstacle avoidance; a handoff from the Arduino to the Jetson Nano via serial to begin fall-detection analysis; and a closed loop back from the Jetson's FALL_DETECTED signal into the Arduino's alert handling. The keypad's panic button was integrated as a second, independent path into the same alert state, confirming the state machine correctly handles multiple concurrent input sources. The browser-based remote-drive interface was also integrated through the same serial command set used by autonomous navigation, with obstacle-safety checks enforced identically in both manual and autonomous modes.

Phase 1 explicitly excludes the following, which are scoped for Phase 2: final mounting and wiring of the GSM module into the assembled unit (SMS alerting is currently validated as a standalone AT-command test and stubs to serial output within the main firmware); and continuous on-robot execution of the Jetson Nano fall-detection pipeline, which has so far been validated against a trained classifier and recorded video rather than as a live process running on the physical robot.

Two of the three items originally scoped for Phase 2 have since been completed. GSM alerting is now implemented end-to-end on replacement hardware (see GSM Integration), and the Jetson Nano fall-detection pipeline now runs continuously on the physical robot via a dedicated perception node, gated on active person-tracking, rather than only against pre-recorded video.

4.7). Health Management Implementation

Medication and health-check confirmation uses the MFRC522 RFID reader over the Arduino's hardware SPI bus. When a tag is presented, checkRFID() reads the card's unique identifier and logs it alongside a timestamp sourced from the DS3231 real-time clock, which retains accurate time across power loss. This RFID-plus-RTC pairing provides the foundation for confirming that a scheduled medication or check-in event was physically acknowledged by the resident, rather than merely displayed and ignored.

The current implementation covers tag detection, UID logging, and timestamping. The scheduling layer described in the system design — four daily medication reminders via buzzer and LCD, and an automated SMS alert to family if a reminder is not confirmed within 15 minutes — is not yet implemented in firmware and is scoped for the next development phase, once reminder scheduling logic is added on top of the existing RTC and RFID foundation.

4.8). Connection Features

The keypad, already integrated for general input, doubles as the one-touch family-alert mechanism specified in the design requirements: pressing # immediately transitions the state machine into ALERT from any state, triggering the buzzer pattern and SMS-alert logic regardless of what the robot was doing at the time. This reuses existing hardware rather than requiring a dedicated panic button. Pressing * while an alert is active acknowledges and silences it, transitioning the system to RESET_STATE.

A separate connection path is provided through the browser-based remote interface (web_interface_for_remote_control/remote_control.py), which streams a live MJPEG camera feed and exposes a D-pad control panel. Drive commands (MOVE_FORWARD,

MOVE_BACKWARD, TURN_LEFT, TURN_RIGHT, STOP) are sent over the same serial command protocol used internally by the Arduino's autonomous logic, and are subject to the same obstacle-safety checks — allowing a family member to view and drive the robot remotely without bypassing collision avoidance.

4.9). Companion Features and Full Integration

Companion behavior centers on a rotating set of eight greeting messages, randomly selected on each PIR trigger with a repeat-rejection check ensuring no two consecutive greetings are identical. Each greeting is displayed on the LCD and paired with a servo wave gesture, sweeping through 90°→150°→30°→90°. The servo is attached only for the duration of the gesture and detached immediately afterward, both to conserve power and to avoid interfering with the motor PWM timers during normal operation.

During IDLE, an ambient LED pulse — driven by a sine-wave brightness function rather than a simple on/off blink — gives the robot a passive, "breathing" visual presence even when not actively interacting with anyone. Together, the greeting rotation, gesture, and idle pulse represent the initial integration of CARA's social-presence behaviours alongside its safety-critical systems, directly supporting the design goal of the robot becoming a familiar, non-clinical presence in the home rather than a purely functional monitoring device.

5). Chapter: Testing and Evaluation

5.1). Evaluation Methodology

Testing followed a bottom-up approach: each hardware module was first validated in isolation before integration, followed by incremental combination testing, and finally whole-system state-machine testing.

Unit-level testing. Each sensor and actuator (RFID reader, ultrasonic ring, PIR sensors, buzzer, LED, LCD, RTC, keypad) was wired and tested independently on a breadboard with a minimal test sketch confirming correct pin mapping and expected sensor output, before being combined into the main firmware. This isolates hardware faults from integration or software-logic faults.

Incremental integration testing. Modules were added to the firmware step-by-step rather than all at once — for example, confirming ultrasonic obstacle detection on the assembled chassis before adding PIR-triggered state transitions, then layering in the remaining sensors — so that any regression could be attributed to the most recently added component.

Systematic fault diagnosis. Where a module failed after integration (for example, one side of the drive motors not responding), diagnosis followed a fixed elimination order: confirm power supply voltage at the component, confirm the correct microcontroller pin is toggling as expected, confirm the driver/interface board is functioning, and finally inspect physical wiring — narrowing the fault to its actual source (in that case, a loose connection) rather than guessing.

Software/ML evaluation. The fall-detection classifier was evaluated using a held-out test split combined with k-fold cross-validation, to check that reported accuracy was not an artefact of a favourable single split. Live camera-based detection was additionally validated against pre-recorded video where live conditions (lighting, availability of hardware) were not controllable, ensuring the detection pipeline was tested under repeatable conditions.

Remote interface testing. The browser-based monitoring and remote-drive interface was tested first over a local shared network before evaluating wider remote-access methods, isolating interface/functionality issues from network-access configuration issues.

5.2). Individual Feature Results

Fall detection classifier. The trained RandomForestClassifier was evaluated on a held-out 20% test split from 644 labelled pose samples (303 fall, 341 not-fall), with 5-fold cross-validation used to check for overfitting. Results: **87.6%** test accuracy, **89.6%** cross-validated accuracy (±4.8%), 84.6% precision and 90.2% recall on the fall class. The confusion matrix (55 true positives, 6 false negatives, 10 false positives, 58 true negatives) shows the classifier is more conservative on the not-fall class than the fall class — consistent with the design decision to prioritise recall over precision, since an undetected real fall is a more serious failure mode than an occasional false alert for an elderly-care application.

GSM communication (SIM800L). AT-command layer testing confirmed the module correctly handshakes, accepts configuration commands, and reports supply voltage. Network registration testing, however, consistently failed despite correct sequencing and an automatic radio-reset recovery routine, which was later diagnosed as a network incompatibility — UAE carriers have discontinued 2G service, and the SIM800L is 2G-only. This is reported here as a negative but informative result: it validated the communication protocol implementation while identifying a hardware selection error that informed us of the subsequent decision to migrate to a SIM7600 module.

RFID access control and patient identification. The RFID reader, keypad, and LCD were validated both individually and combined in a single test sketch, correctly distinguishing an authorised confirmation tag from other tags/cards and reflecting scan and keypress events on the display within the same control loop. Beyond RFID alone, a photo-enrolment and face-recognition capability was added and tested: the system can be trained to recognise a specific enrolled individual via either the physical keypad or the browser-based interface, with both trigger paths converging on identical underlying capture/train logic.

Obstacle avoidance and drive control. An escalating-angle scan-and-avoid strategy was implemented and bench-validated on the assembled chassis, reliably locating a clear heading or correctly reporting itself fully enclosed. A hardware constraint was identified during this testing: the originally fitted drive motors could not sustain adequate torque at the low PWM duty cycles required for slow, elderly-care-appropriate motion, traced through torque/RPM calculation to a gear-ratio mismatch rather than a control-logic fault; replacement motors were specified accordingly.

Display and real-time-clock hardware fault diagnosis. A persistent display fault was traced, through direct voltage-level testing of the microcontroller's clock pin independent of any library code, to physical damage on that specific pin — confirmed further by the same display and real-time-clock modules functioning correctly when tested on a separate microcontroller. The display was recovered by re-implementing its communication protocol at a lower level on an undamaged pin; the real-time clock, which depends on the same damaged hardware interface, remains a known outstanding limitation.

5.3). System Integration Evaluation

Beyond the individual module tests described in the Evaluation Methodology, three combined-system tests were conducted to validate cross-subsystem behaviour. A four-module integration test (LCD, RFID access control, buzzer, and keypad) confirmed that these peripherals could run concurrently on the Arduino Mega without pin conflicts or timing interference, and that a keypad-triggered event (tag scan, key press) could be reflected correctly on the LCD within the same control loop. A second integration point validated the ROS2-based perception pipeline on the Jetson Nano: the person-tracking node (face-recognition primary, HSV colour-band fallback) and the fall-detection node were connected via a shared lock-state topic, so that fall detection only executes when the system is actively tracking the enrolled patient, rather than analysing any person visible to the camera. This was confirmed by observing the fall-detection status transition to a distinct "no target" state whenever the tracking lock was lost, and resume active monitoring immediately once the lock was re-acquired. A third integration point validated the browser-based remote interface end-to-end: drive commands issued from a phone browser were shown to reach the Arduino's obstacle-checked motor-control logic through the same serial command path used by the autonomous state machine, and patient-enrolment actions (photo capture, model training) could be triggered equivalently from either the physical keypad or the same web interface, both converging on identical enrolment logic.

5.4). Objectives Achievement

Objective 1 — navigation and obstacle avoidance. An escalating-angle obstacle-avoidance strategy was implemented and bench-tested on the assembled chassis: on detecting an obstacle within the stop threshold, the robot attempts a scan sequence (45°, 90°, 105°… up to 180°) alternating right and left until a clear heading is found. This logic was validated qualitatively across repeated bench trials and reliably found an opening or correctly reported itself fully enclosed. Formal collision-free-trial statistics against the 80% target could not be completed in this phase, however, because the originally fitted drive motors were found to lack sufficient torque at the low PWM duty cycles required for the smooth, low-speed motion appropriate to an elderly-care context — a mechanical constraint identified through systematic torque/RPM calculation from the chassis mass, wheel diameter, and target speed, rather than a control-logic fault. Replacement gear motors were specified against these calculations; this objective is assessed as logic-complete and partially validated, pending replacement-motor installation and a formal trial count.

Objective 2 — AI fall detection. Achieved and exceeded on the held-out evaluation: 87.6% test accuracy and 89.6% cross-validated accuracy against an 80% target, with recall (90.2%) deliberately prioritised over precision. Beyond the original design, fall detection was further constrained during this project to run only while the perception pipeline is actively tracking the enrolled patient specifically, rather than analysing any person in frame — closing a real gap where a visitor or bystander could otherwise have triggered a false alert. The detector now also runs live against the on-robot camera feed via a ROS2 node, rather than only against pre-recorded video as in the original implementation.

Objective 3 — GSM emergency alerting. The originally specified SIM800L module was proven unable to meet this objective for reasons independent of implementation: UAE mobile carriers have discontinued 2G service, and the SIM800L is 2G-only, making network registration structurally impossible regardless of wiring or power correction — a conclusion reached only after systematically eliminating SIM-specific and power-supply causes first. Following this finding, an A76xx-series LTE Cat-1 module was procured, and a complete interactive SMS system was implemented and bench-validated: outbound messages, delivery-report confirmation, and inbound message logging were all demonstrated working. This objective is assessed as achieved on the replacement hardware.

Objective 4 — health management. RFID-based confirmation is implemented and reliably functional, extended during this project into two complementary confirmation paths (RFID tag or keypad acknowledgement). Photo-based patient enrolment was also added beyond the original design. Scheduled reminder timing and automated family SMS on a missed confirmation remain implemented only as a fixed-delay test trigger rather than full time-of-day scheduling, pending a still-outstanding real-time-clock fix. This objective is assessed as partially achieved.

Objective 5 — complete system evaluation. Individual features have each been validated in isolation or in small combined groups. A single continuous end-to-end trial exercising all fifteen features together in one uninterrupted session has not yet been conducted, and is identified as the immediate next evaluation step.

5.5). Limitations and Threats to Validity

Single-subject and laboratory-only testing. All trials, including fall-detection classifier data collection and patient-enrolment testing, used the researcher as the sole test subject in a controlled laboratory environment, consistent with the declared scope of this dissertation. Results should not be generalised to elderly users, whose gait, posture baseline, and interaction patterns with the keypad/RFID interface may differ meaningfully; formal ethical clearance for elderly-participant testing was not sought for this phase.

Hardware faults encountered mid-project. Two components failed during implementation in ways that could not be attributed to wiring or code on first inspection: an RFID reader that communicated correctly at the register level but could not energise a passive tag (attributed to a defective unit, confirmed by cross-testing an identical wiring pattern on a second reader and a second microcontroller), and an LCD hang traced to a physically damaged microcontroller pin (confirmed via direct voltage-level probing independent of any library code, corroborated by the same LCD functioning correctly on a separate microcontroller). Both required a systematic elimination process — isolating power, wiring, module, and microcontroller as independent variables — and both were resolved by replacement or a lower-level workaround. These faults are reported here for methodological transparency rather than omitted.

Unvalidated fallback conditions. The colour-band fallback for person-tracking was implemented and its control logic exercised, but the colour-detection thresholds remain an untuned placeholder pending a finished physical wristband; its real-world reliability under the intended failure condition (poor facial visibility) has not yet been empirically measured. Pose-based fall detection while the system is band-locked rather than face-locked has also not been separately validated.

Single-identity recognition testing. Patient-identification testing to date has validated that the system can be trained to recognise one enrolled identity and distinguish it from a small number of other individuals encountered incidentally; it has not been tested under a controlled multi-person discrimination protocol.

6). Chapter: Conclusion and Future Work

6.1). Summary of Research

This dissertation set out to design, build, and evaluate CARA, a low-cost companion assistive robot combining an Arduino Mega for real-time sensing and control, an NVIDIA Jetson Nano for on-device AI perception, and cellular (GSM/LTE) communication for connectivity-independent family alerting. Across the implementation phase, fifteen planned features were developed to varying degrees of completeness, spanning autonomous navigation, AI-based fall detection, emergency communication, health management, remote family connection, and companion-style interaction. Beyond the originally scoped design, the implementation process itself surfaced and resolved several non-trivial engineering problems — a cellular technology mismatch with local network infrastructure, two independent hardware-component failures, and an underpowered drive system — each diagnosed through a systematic, evidence-based elimination process rather than assumption, and each documented as part of this research's account of building assistive technology under real cost and component-availability constraints.

6.2). Objectives Achievement

Of the five stated objectives, two are assessed as fully achieved (AI fall detection, exceeding its 80% accuracy target; GSM emergency alerting, achieved on replacement hardware after the originally specified module was shown to be structurally incompatible with local network infrastructure), two as partially achieved (autonomous navigation, logic-complete but constrained by an underpowered drive system pending replacement; health management, RFID confirmation and patient enrolment complete but time-of-day scheduling outstanding), and one as in progress (a single continuous end-to-end evaluation of all fifteen features together has not yet been conducted). Detailed evidence for each is presented in Chapter 5.

6.3). Research Contribution

This research contributes in four ways beyond the working prototype itself. First, a systematic integration framework is demonstrated for combining a microcontroller-based real-time control layer with a separate AI perception layer, coordinated over a simple serial command protocol, that keeps safety-critical logic (obstacle avoidance, alert handling) on the deterministic microcontroller regardless of which higher-level system requested the action — evidenced by remote-drive and autonomous-drive commands being subject to identical collision checks. Second, a hybrid person-identification and tracking architecture is contributed: primary identification by trained face recognition, with an explicit, engineered fallback to a physical marker (colour band) precisely when the primary method is expected to degrade, and a defined re-acquisition behaviour that uses the fallback's last-known position to narrow the search area for recovering face-based tracking — coupling fall-detection execution to this tracking lock so that safety monitoring is scoped to a specific, enrolled individual rather than any person in view. Third, empirical guidance is provided for fall-detection classifier design on constrained embedded hardware, including a concrete two-layer confirmation strategy that balances detection latency against susceptibility to single-frame pose noise, together with a worked, real accuracy/precision/recall result set. Fourth, this dissertation documents a transferable methodology for diagnosing hardware faults in low-cost embedded systems under field conditions — including register-level protocol probing, raw digital-signal-level testing independent of library code, and cross-device isolation testing — offered as a practical template for other resource-constrained hardware projects rather than only reporting the faults' resolutions.

6.4). Limitations

The principal limitations of this work are its single-subject, laboratory-only testing scope; a drive-motor hardware constraint that prevented formal navigation trials against the stated collision-free target; two mid-project hardware faults (RFID and LCD/RTC) that consumed development time better spent on feature completion; an incomplete health-management scheduling feature; and an untuned physical fallback (the colour band) for person-tracking. A full discussion with supporting evidence is given in Chapter 5.

6.5). Future Work

Immediate next steps follow directly from the limitations identified above. The replacement drive motors specified via torque/RPM calculation should be installed and the navigation objective's collision-free trial count completed against the original 80% target. A physical wristband should be fabricated and its colour-detection thresholds tuned under real lighting conditions, followed by a dedicated evaluation of the band-locked fall-detection condition specifically. Medication-reminder scheduling should be completed against real time-of-day rather than a fixed test delay, which in turn requires resolving the real-time-clock module's current hardware fault. A single continuous end-to-end trial exercising all fifteen features in one session should be conducted to complete Objective 5. Beyond these near-term items, securing ethical clearance for testing with elderly participants, extending trials beyond the laboratory into a representative home environment, and formally re-evaluating the fall-detection classifier's accuracy under those less controlled conditions represent the clearest path toward the commercial-grade reliability this dissertation identifies as currently unmet.

6.6). Final Reflection

Undertaking this dissertation meant working through problems that a purely theoretical design would not have surfaced: a cellular module that was electrically functional but structurally unable to register on a discontinued network generation; a reader module that answered every register query correctly yet still could not power a tag; a display that hung the entire control loop because of a single damaged pin rather than any line of code. Each of these required setting aside the assumption that a fault must be a software bug, and instead building a habit of isolating variables — power, wiring, the specific component, the specific pin — one at a time until the actual cause could be shown, not guessed. That discipline, more than any single feature, is the practical skill this project leaves me with, and it shaped how the whole system was eventually integrated: safety and control logic kept deliberately simple and deterministic on the microcontroller, with the more experimental AI perception work kept separate and never allowed to bypass it.

References

[1] U. Nations, "World Population Ageing 2020 Highlights," United Nations Department of Economic and Social Affairs, New York, 2020.

[2] W. H. Organization, "Ageing with Dignity: The Importance of Strengthening Care and Support Systems for Older Persons Worldwide," WHO Sri Lanka, Colombo, 2024.

[3] A. T. P. L. Abeykoon and R. Elwalagedara, "Emerging Social Issues of Population Ageing in Sri Lanka," Institute for Health Policy, Colombo, 2007.

[4] Dailymirror, "Challenges Faced by Elders in Sri Lanka," Dailymirror, 08 October 2016. [Online]. Available: https://www.dailymirror.lk/print/opinion/Challenges-faced-by-Elders-in-Sri-Lanka/172-117076?utm_source=chatgpt.com. [Accessed 06 March 2026].

[5] W. H. Organization, "Global Report on Falls Prevention in Older Age," World Health Organization, Geneva, 2007.

[6] M. Firdhous and P. M. Karunaratne, "An ICT Enhanced Life Quality for the Elderly in Developing Countries: Analysis Study Applied to Sri Lanka," *arXiv,* 2012.

[7] E. Broadbent, R. Stafford and B. MacDonald, "Acceptance of healthcare robots for the older population: Review and future directions," *International Journal of Social Robotics,* 2009.

[8] G. Demiris and H. Hensel, "Technologies for an aging society: A systematic review of smart home and telehealth applications," *Journal of the American Medical Informatics Association,* vol. 15, no. 1, p. 42–55, 2008.

[9] R. Beer, J. Prakash, B. Mitzner and W. Rogers, "Understanding challenges in the design of assistive technology for low-resource settings," *Gerontechnology,* 2012.

[10] P. K. Dautenhahn, "Socially intelligent robots: Dimensions of human–robot interaction," *Philosophical Transactions of the Royal Society A,* vol. 362, p. 679–704, 2004.

[11] M. Firdhous and P. M. Karunaratne, "An ICT Enhanced Life Quality for the Elderly in Developing Countries: Analysis Study Applied to Sri Lanka," *arXiv,* 2012.

[12] W. H. Organization, "Ageing with dignity: Strengthening care and support systems for older persons worldwide," WHO, Colombo, 2024.

[13] S. Wijesinghe, "Addressing the urgency for healthy aging in Sri Lanka," *WHO South-East Asia Journal of Public Health,* 2025.

[14] World_Bank, "Enhancing Skills in Sri Lanka for Inclusion, Recovery, and Resilience," World Bank, Washington, DC, 2024.

[15] E. NewsNow, "Ageing Population Increased from 12% to 18%: Sri Lanka Among Fastest-Ageing Nations in Asia," 2025. [Online]. Available: https://english.newsnow.lk/2025/10/31/ageing-population-increased-from-12-to-18-sri-lanka-among-fastest-ageing-nations-in-asia/?utm_source=chatgpt.com. [Accessed 07 March 2026].

[16] R. P. Subadarshini, "Embracing the opportunities and addressing the challenges of the aging population," 23 February 2024. [Online]. Available: https://ceylontoday.lk/2024/02/23/embracing-the-opportunities-and-addressing-the-challenges-of-the-aging-population/?utm_source=chatgpt.com. [Accessed 07 March 2026].

[17] T. Shibata and K. Wada, "Robot therapy: A new approach for mental healthcare of the elderly — A case study with PARO," *International Journal of Social Robotics,* vol. 57, p. 378–386, 2011.

[18] K. Wada and T. Shibata, "Living with seal robots – Its sociopsychological and physiological influences on the elderly at a care house," *IEEE Transactions on Robotics,* vol. 23, no. 5, p. 972–980, 2007.

[19] R. Kittmann, T. Fröhlich, J. Schäfer, U. Reiser, F. Weißhardt and A. Haug, "Let me Introduce Myself: I am Care-O-bot 4, a Gentleman Robot," 2015.

[20] M. Tonkin, J. Vitale, S. Herse, M. Williams, W. Judge and X. Wang, "Design methodology for the UX of HRI: a field study of a commercial social robot at an airport," New York, NY, USA, 2018.

[21] R. Bemelmans, A. Gelderblom, R. Jonker and H. de Witte, "Socially assistive robots in elderly care: A systematic review into effects and effectiveness," *Journal of the American Medical Directors Association,* vol. 12, no. 2, p. 114–120, 2012.

[22] E. Broadbent, R. Stafford and B. MacDonald, "Acceptance of healthcare robots for the older population: Review and future directions," *International Journal of Social Robotics,* vol. 1, no. 4, p. 319–330, 2009.

[23] A. K. Bourke, J. V. O'Brien and G. M. Lyons, "Evaluation of a threshold-based tri-axial accelerometer fall detection algorithm," *Gait & Posture,* vol. 26, no. 2, p. 194–199, 2007.

[24] C. Rougier, J. Meunier, A. St-Arnaud and J. Rousseau, "Robust video surveillance for fall detection based on human shape deformation," *IEEE Transactions on Circuits and Systems for Video Technology,* vol. 21, no. 5, p. 611–622, 2011.

[25] H. Rimminen, J. Lindström, M. Linnavuo and R. Sepponen, "Detection of falls among the elderly by a floor sensor using the electric near field," *IEEE Transactions on Information Technology in Biomedicine,* vol. 14, no. 6, p. 1475–1476, 2010.

[26] S. Saraswat and G. Malathi, "Pose Estimation based Fall Detection system using MediaPipe," 2024.

[27] P. Gorce and J. Jacquier-Bret, "Fall Detection in Elderly People: A Systematic Review of Ambient Assisted Living and Smart Home-Related Technology Performance," *Sensors,* vol. 25, no. 21, p. 6540, 2025.

[28] T. Alhmiedat, A. M. Marei, W. Messoudi, S. Albelwi, A. Bushnag, Z. Bassfar and A. O. … Elfaki, "A SLAM-Based Localization and Navigation System for Social Robots: The Pepper Robot Case," *Machines,* vol. 11, no. 2, 2023.

[29] J. Zhao, S. Liu and J. Li, "Research and Implementation of Autonomous Navigation for Mobile Robots Based on SLAM Algorithm under ROS," *Sensors,* vol. 22, no. 11, 2022.

[30] B. M. F. d. Silva, R. S. Xavier and L. M. G. Gonçalves, "Mapping and Navigation for Indoor Robots under ROS: An Experimental Analysis," *Preprints,* 2019.

[31] R. Alqobali, M. Alshmrani, R. Alnasser, A. Rashidi, T. Alhmiedat and O. M. Alia, "A Survey on Robot Semantic Navigation Systems for Indoor Environments," *Applied Sciences,* vol. 14, no. 1, 2024.

[32] A. Eirale, M. Martini and M. Chiaberge, "RL-DWA Omnidirectional Motion Planning for Person Following in Domestic Assistance and Monitoring," *arXiv,* 2022.

[33] M. Arun, K. Dinesh, M. Sarumathan and R. Santhoshi, "IoT-Based Patient Health and Fall Monitoring System with GSM Alerts," *International Journal for Research in Applied Science and Engineering Technology (IJRASET),* 2025.

[34] G. M. Hassan, "A Real-Time Monitoring System for the Elderly Based on the GSM Network," *Journal La Multiapp,* vol. 3, no. 6, p. 287–293, 2023.

[35] Y. N. Al Husaini, M. Al Nuaimi and P. C. Sherimon, "IoT Based Remote Health Monitoring System for Patients and Elderly People," *International Journal of Biology and Biomedicine,* vol. 7, no. 2, pp. 21-25, 2022.

[36] P. S. Sritha and R. S. Valarmathi, "A Reliable Remote Health Monitoring System for Elderly People," *International Journal of Scientific & Technology Research (IJSTR),* vol. 9, no. 3, p. 5995–5998, 2020.

[37] M. E. Karar, H. I. Shehata and O. Reyad, "A Survey of IoT-Based Fall Detection for Aiding Elderly Care: Sensors, Methods, Challenges and Future Trends," *Applied Sciences,* vol. 12, no. 7, p. 3276, 2022.

[38] D. Sirizi, M. Sabet, A. Yahyaeian, J. Bacsu, M. Smith and Z. Rahemi, "Evaluating Human-Robot Interactions to Support Healthy Aging-in-Place," *Journal of Applied Gerontology,* vol. 44, no. 2, p. 266–277, 2025.

[39] M. Crawford, J. Smith, K. Jones and e. al., "Social Robot in Home Care: Acceptability and Utility Among Community-Dwelling Older Adults," *Innovation in Aging,* vol. 9, no. 1, 2025.

[40] A. Elsheikh, D. Al-Thani and A. Othman, "Exploring Fear in Human-Robot Interaction: A Scoping Review of Older Adults' Experiences with Social Robots," *Frontiers in Robotics and AI,* vol. 12, 2025.

[41] A. Cruces, A. Jerez, J. Bandera and A. Bandera, "Socially Assistive Robots in Smart Environments to Attend Elderly People — A Survey," *Applied Sciences,* vol. 14, no. 23, p. 11352, 2024.

[42] S. M. T. U. Raju, "Enhancing Human-Robot Interaction in Healthcare: A Study on Nonverbal Communication Cues and Trust Dynamics with NAO Robot Caregivers," arXiv, 2025.

[43] A. Demofonti, G. Carpino, L. Zollo and M. Johnson, "Affordable Robotics for Upper Limb Stroke Rehabilitation in Developing Countries: A Systematic Review," *IEEE Transactions on Medical Robotics and Bionics,* vol. 3, no. 1, p. 33–45, 2021.

[44] W. Han, L. Liu and W. Lai, "Smart eldercare robotics and disability: barriers, risks, and policy gaps in the Chinese context," *Disability and Society,* pp. 1-6, 2025.

[45] S. Almuaythir, A. Singh and M. Alhusban, "Robotics technology: catalyst for sustainable development—impact on innovation, healthcare, inequality, and economic growth," *Discover Sustainability,* vol. 5, p. 486, 2024.

[46] J. Favela, D. Cruz-Sandoval, M. da Rocha and D. Muchaluat-Saade, "Social Robots for Healthcare and Education in Latin America," *Communications of the ACM,* vol. 67, no. 8, pp. 70-71, 2024.

[47] Wikipedia, "WeRobotics," 30 November 2025. [Online]. Available: https://en.wikipedia.org/wiki/WeRobotics. [Accessed 07 March 2026].

[48] J. Kim, S. Kim and G. Lee, "Development of an IoT-based Smart Pillbox System for Elderly Medication Management," *IEEE Access,* vol. 10, 2022.

[49] M. Alshamman and S. Kapetanakis, "Medication Adherence Model Using AI and Mobile Notifications for Elderly Patients," *IEEE Journal of Biomedical and Health Informatics,* vol. 27, no. 4, 2023.

[50] A. Sheikh and A. Khalid, "RFID-Based Confirmation System for Medication Adherence in Elderly Patients," *Sensors (MDPI),* vol. 24, no. 3, 2024.

[51] A. Saxena and O. Sharma, "Design and Evaluation of a Smart Medication Dispenser with Reminder and Alert System," 2023.

[52] H. Lee and J. Park, "Wearable IoT Framework for Medication Monitoring and Reminder System," *IEEE Internet of Things Journal,* vol. 11, no. 8, 2024.

[53] S. Kim, S. Park, H. Hwang, S. Moon and J. Park, "Effectiveness of Mobile Health Intervention in Medication Adherence: a Systematic Review and Meta-Analysis," *J Med Syst,* vol. 49, no. 1, p. 13, 2025.

Appendix A) Circuit Wiring Diagram

Table 8 Circuit Wiring Diagram (Author developed)

Subsystem	Pins
Motors (L298N)	IN1=22, IN2=23, IN3=24, IN4=25; ENA=11 (PWM), ENB=12 (PWM)
Ultrasonic ring (×6)	FL=26/27, FR=28/29, BL=30/31, BR=32/33, L=34/35, R=36/37
PIR	Front=2, Rear=3
Buzzer / LED	Buzzer=8, LED=10
Servo (MG90S)	Pin 46
LCD (bit-banged I2C)	SDA=20, SCL=A0
RTC	SDA=20, SCL=21 (shared hardware bus — see Limitations)
RFID (SPI)	SS=53, RST=49 (SCK=52, MOSI=51, MISO=50 fixed)
Keypad (4×4)	Rows=38,39,40,41; Cols=42,43,4,5
GSM (A7670C)	Hardware Serial1: TX=18, RX=19; PWRKEY=9

Appendix B) Arduino Firmware Listing

The firmware evolved across the project; cara_main_jetson.ino, listed in full below, is the current, actively-used firmware — rebuilt to work around the damaged I2C pin, extended with ramped/speed-controllable motor drive, the serial command set, and keypad-triggered patient enrolment."

Then paste in experiments/Main_Codes/Arduino/cara_main_jetson/cara_main_jetson.ino

Appendix C) Python Fall Detection Code

The fall-detection implementation consists of two files: the core feature-extraction/classifier logic, and the ROS2 node that wraps it for live on-robot execution, gated on the person-tracker's lock state."

Paste fall_detection.py then fall_detection_node.py (both in experiments/)

Appendix D) Raw Test Results

	Predicted Fall	Predicted Not Fall
Actual Fall	55 (TP)	6 (FN)

Actual Not Fall	10 (FP)	58 (TN)

Test accuracy: 87.6% · Cross-validated: 89.6% (±4.8%) · Precision: 84.6% · Recall: 90.2%

Appendix E) AI Usage Statement

In accordance with Middlesex University's policy on the use of artificial intelligence in academic work, this statement discloses how AI tools were used during this dissertation. An AI coding assistant was used throughout the implementation phase to draft and iterate Arduino/C++ and Python code based on requirements and test results supplied by the author, to assist with systematic hardware-fault diagnosis by proposing isolation tests, and to help structure and draft sections of this report based on the author's project history and stated experimental outcomes.

All engineering decisions, hardware selection, wiring, test design, and interpretation of results were made by the author. All physical construction and hands-on testing were performed by the author. All data reported in Chapter 5 were generated by the author's own test execution. Where AI-drafted text is included, it was reviewed, edited, and approved by the author, who takes full responsibility for its accuracy.